

Protective effect of fruits, vegetables and the Mediterranean diet on asthma and allergies among children in Crete

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Background Atopy is not uncommon among children living in rural Crete, but wheeze and rhinitis are rare. A study was undertaken to examine whether this discrepancy could be attributed to a high consumption of fresh fruit and vegetables or adherence to a traditional Mediterranean diet. Methods A cross-sectional survey was performed in 690 children aged 7–18 years in rural Crete. Parents completed a questionnaire on their child's respiratory and allergic symptoms and a 58-item food frequency questionnaire. Adherence to a Mediterranean diet was measured using a scale with 12 dietary items. Children underwent skin prick tests with 10 common aeroallergens. Results 80% of children ate fresh fruit (and 68% vegetables) at least twice a day. The intake of grapes, oranges, apples, and fresh tomatoes—the main local products in Crete—had no association with atopy but was protective for wheezing and rhinitis. A high consumption of nuts was found to be inversely associated with wheezing (OR 0.46; 95% CI 0.20 to 0.98), whereas margarine increased the risk of both wheeze (OR 2.19; 95% CI 1.01 to 4.82) and allergic rhinitis (OR 2.10; 95% CI 1.31 to 3.37). A high level of adherence to the Mediterranean diet was protective for allergic rhinitis (OR 0.34; 95% CI 0.18 to 0.64) while a more modest protection was observed for wheezing and atopy. Conclusion The results of this study suggest a beneficial effect of commonly consumed fruits, vegetables and nuts, and of a high adherence to a traditional Mediterranean diet during childhood on symptoms of asthma and rhinitis. Diet may explain the relative lack of allergic symptoms in this population.